

JCP campaign notebook

Continuous coordinate-network bases for Petrov–Galerkin reduced models

Tunable-NM-ROM project

Rebuilt October 8, 2026

What this is. The working notebook for the JCP paper: one entry per update, newest last, built by `notebook/build.sh`. It records decisions, designs, results (generated, never typed), and anything retracted. The canonical chronology stays in `LAB-LOG.md`; the plan is `reports/2026-10-06-jcp-offmesh-paper-plan.md`; the outline is `reports/2026-10-06-jcp-combined-paper-outline/`.

Contents

1	2026-10-06 — Campaign start and Phase 0	2
2	2026-10-08 — Launch: accuracy lanes (C) and two mechanism checks (A)	5

1 2026-10-06 — Campaign start and Phase 0

Thesis

A reduced trial basis that is a *continuous, importance-ordered* function of position, $u(x) = G(x)c$, gives two properties. **Ordered:** one trained model offers an accuracy/cost dial at deployment (width R' , head on or off). **Continuous:** the nonlinear term is an integral, so it can be hyper-reduced by classical quadrature (tensor Gauss, rank-1 lattices) at points that ignore the mesh, instead of quadrature fitted to snapshots on the mesh. The paper combines the discarded ICLR material (the dial) with Hari's off-mesh quadrature study. The ICLR paper will not be submitted.

Decisions (user, 2026-10-06)

topic	decision
Paper	One combined JCP paper; ICLR dropped.
Off-mesh code	Copied byte-identically into each lane's <code>vendor/</code> with <code>PROVENANCE.json</code> (not forked from the quadrature branches).
Full-order comparators	Run <i>and compare against</i> all three of coarse-grid, second-order and spectral solvers, in addition to Newton-BiCGStab/GMRES and IMEX-CG. Supersedes the 2026-09-21 “named iterative only” rule.
Examples	Keep the current PDEs and add harder ones (chosen in the lane designs).
NS 3D	Keep the lane and keep improving its coordinate-network bank.
Worktrees	Claude names and creates them without asking (now in <code>CLAUDE.md</code> and <code>AGENTS.md</code>).
Notebook	This LaTeX notebook, rebuilt after every update.

Phase 0 checklist

- [\[done\]](#) Plan and five planning notes, revised after a Codex audit: `reports/2026-10-06-jcp-offmesh-paper-plan.md`.
- [\[done\]](#) Paper outline (PDF): `reports/2026-10-06-jcp-combined-paper-outline/`.
- [\[done\]](#) Literature check of the unread preprints and a 2024–2026 sweep (plan appendix 07); see below.
- [\[done\]](#) Week-1 lanes created (table below).
- [\[done\]](#) `CLAUDE.md`, `AGENTS.md` and memory updated with the decisions.
- [\[user\]](#) Withdraw the ICLR submission on OpenReview if it is still active.
- [\[user\]](#) Agree authorship and the split of work with Hari; get his consent to use his code and results.

Literature check: outcome

None of arXiv 2505.14595 (Φ -ROM, a grid-based contrast), 2507.07830 (meshless SPH projection, no hyper-reduction), 2508.21279 (energy-conservative EQP, a fitted method) or the LiCROM cubature (random mesh vertices, equal weights) pre-empt the claims. No ROM paper found uses weighted classical quadrature for the reduced residual. Per contribution:

- ordered tunable bank: **novel for ROMs**; credit nested dropout (Rippel et al. 2014), slimmable networks, POD truncation, LiCROM;
- classical quadrature replacing fitted hyper-reduction: **novel as a combination**; ingredients partly pre-empted (Monte Carlo/uniform points in Simplicits, SNF-ROM, Weder et al.; lattices in PINN training; fitted off-mesh points in CECM);
- decoupling from the mesh’s $O(h)$ error: **partly pre-empted** (SNF-ROM has the mechanism); Ingimarson–Rebholz–Iliescu (CMAME 2022) argue the opposite position and must be addressed;
- rule-family behaviour on tensor sine tests: **novel in the ROM setting**, as known theory with a new consequence;
- data-free error indicator: **partly pre-empted in principle**; call it an indicator, never a certificate.

Still to read before submission: arXiv 2607.27471, 2607.10965, 2602.01397, 2504.15657.

Week-1 lanes

All fork from `exp/2026-10-01-ns3d-coordnet-bank @ 21c175a1b`; each mirrors to `origin/codeonly/exp/<lane>` with its own `sync.github.sh`, and vendors 69 files of off-mesh code, each checked blob by blob against its source commit (`exp/2026-10-01-quadrature-study @ 84d28e303`, `exp/2026-10-01-quadrature-burgers3d @ aab6b6927`).

lane (<code>worktrees/</code> , branch <code>exp/...</code>)	experiments	role
2026-10-06-jcp-references	E0	order-verified second-order references; pseudo-spectral cross-check; sealed and out-of-distribution cohorts in a read-only store
2026-10-06-jcp-mechanism	E1, E2	rule ladders against theory; second-order stencil tensor, mesh-node analytic gradient, coarse-trained bank, all-continuum arm
2026-10-06-jcp-fom-pareto	E4	full-order cost ladders (iterative, IMEX-CG, second-order, coarse-grid, spectral); timing harness
2026-10-06-jcp-hr-baselines	E5	POD-LSPG/Galerkin with ECSW, Q-DEIM, GNAT; POD + exact tensor; best-practice EQ; CROM-style sampler

Harder examples: candidates for the designs

- time-dependent reaction–diffusion with e^u (Frank-Kamenetskii): non-polynomial, so no tensor exists;
- porous-medium diffusion $\nabla \cdot (u^m \nabla u)$ with non-integer m ;
- Burgers at lower viscosity / narrower bumps, to find where the rule size and bank floor break;
- a non-box domain (e.g. L-shape with a distance-function Dirichlet factor and a composite rule);
- NS 3D with an improved coordinate-network bank.

Carried-forward corrections (from the Codex audit of the plan)

Retracted: the explicit Gauss strip-width bound in planning note 03; “lattices converge like m^{-3} ” as an asymptotic rate (only a conditional upper bound); Hari’s “super-algebraic lattice” and “tent adds a kink” statements for this integrand; the per-lane sealing of test cohorts (now global); the 5 % timing fault under a 10 % gate; Gauss “exactness” on sines; the rule-doubling “certificate”; the 5–7-day schedule (at least 7.6 days on four GPUs).

Next (Phase 1)

For each week-1 lane: write `DESIGN.md` → Codex design audit → code → Codex code audit → real-size smoke with every control firing → cluster jobs → Codex results audit → generated report and a notebook entry. Lanes run as separate subagents, one worktree each, with distinct cluster namespaces.

2 2026-10-08 — Launch: accuracy lanes (C) and two mechanism checks (A)

Decision (user)

Push accuracy first: run the three Group C directions, and from Group A only the two most important checks. Every step in every lane is gated by a Codex adversarial check (user instruction); the lane rules are in `reports/2026-10-08-jcp-lane-agent-rules.md`.

Lanes running

lane	question	cluster cap
2026-10-08-jcp-wide-bank (C1)	Now that off-mesh memory is linear in R' , how far does a wider bank push accuracy? 2D at $R' = 512$; trimming the test count M ; training a 3D bank with $R = 1024$; the error/cost scaling law.	1 job
2026-10-08-jcp-time2 (C2)	Does second-order time stepping (Crank–Nicolson, BDF2) remove the backward-Euler error and allow larger steps? Includes a fair second-order full solver.	1 job
2026-10-08-jcp-smooth-bank (C3)	Does a bank trained for its derivatives (Sobolev loss, smoother features) lower the error floor and need fewer quadrature points? Includes a coarse-data control.	1 job
2026-10-06-jcp-mechanism (A1, A2)	Is the gain from leaving the mesh or from the exact gradient (mesh nodes with the analytic gradient)? Does the old error follow the stencil order (upwind vs central gap slope)? A3/A4 deferred.	1 job

Corrections recorded

Retracted: “Our trained bank already has more columns than we used” was wrong for 3D: the 3D bank has exactly 512 columns, all used. A wider 3D bank must be trained; only 2D has headroom (384 of 512 used).

Status

All accuracy numbers from these lanes are *provisional* until the second-order references (E0) exist. The Group B lanes (references, full-solver cost ladder, POD baselines) are not started.